

Year 10 Science
Semester 1 Examination 2018 - Revision

Astronomy

1. Complete the following table.

TERM	DEFINITION
light year	Distance light travels in one year.
constellation	Group of stars forming a pattern.
galaxy	Collection of stars held together by gravity.
nebula	Gas cloud.
supernova	Massive explosion of a massive star near the end of its life.
red giant	Medium stars swell in size as they run out of fuel.
black hole	Very dense and huge gravity. Light can't escape.
sunspot	Dark areas on the surface of the Sun - sites of large eruptions.
corona	Upper atmosphere of the Sun.
fusion	Process producing energy in stars - hydrogen becomes helium.

2. Light travels at 300,000 km/sec. A star is 97.6 light years away. We can only travel at about 15 - 20 km/second with our best rockets.

(a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

YES ☒ NO (Circle your answer.)

Reasons:

- Spaceship is far too slow.
- The distance is huge and will take hundreds of years to reach.
- Will die well before we get there.

(b) Suggest **one way** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

- Develop craft that can travel near the speed of light.
- Freeze people and wake them up hundreds of years later.
- Learn how to warp space and fold it on itself - makes the distance much shorter.

3. (a) What is the major element found in the Sun? hydrogen

(b) What happens to this element so that the Sun produces its energy?

Hydrogen atoms collide and become helium - needs 1 million °C to do this. Energy is released.

(c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

The lost mass becomes the energy that the Sun emits.

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
very hot	blue
cool	red
moderate	yellow

5. (a) About how old is our Sun (in millions of years)? 5000 million years

(b) How much longer will it "live"? 5000 million years

6. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → red giant → nova → white dwarf → black dwarf

(b) Do similar flow diagrams for the following types of stars.

Huge star → red supergiant → supernova → neutron star

Giant star → red supergiant → supernova → black hole

(c) The difference between a **nova** and a **supernova** is the size of the explosion that takes place. What happens to a star to cause it to explode?

Star starts to run out of fuel and (ultimately) collapses on itself.

7. (a) Describe the size of the gravity near a black hole.

huge

(b) Why does a black hole appear black?

Gravity is so big that light can't escape from it.

8. Complete the following matrix to compare the Big Bang and Steady State Theories, using ticks to show which statements apply to one, both or neither theory.

Statement	Big bang theory	Steady state theory
The universe has no beginning.		Y
The universe began with a single point called singularity.	Y	
The universe is expanding.	Y	
The universe always looks the same.		Y
The red shift in the spectrum of visible light coming from stars and galaxies provides evidence for the theory.	Y	
New stars and galaxies are created to replace those that move away due to expansion of the universe.		Y
This theory explains the amount of helium in the universe.	Y	
This theory is supported by the measurement of the current temperature of the universe (about -270°C).	Y	
This theory was first supported by a Catholic priest.	Y	
The theory will never be proven incorrect.		
An end was put to this theory in 1965.		Y

9. Why are the constellations we see now so different from the way they were many centuries ago?

Stars have exploded at various times, so some have disappeared.

10. How have scientists gained their knowledge of the life and death of stars if the processes involved take millions of years to occur?

Light from distant stars takes millions of years to get to us, so we see them as they were at that time. Closer stars appear different, so we can see them at different stages.

11. What is the difference between a neutron star and a black hole?

Neutron star - remains of a massive star; dense; composed of neutrons.

Black hole - remains of a giant star; very dense; huge gravity; light can't escape.

12. What makes stars group together to form galaxies?

Gravity pulls all of the matter and stars together.

13. The Doppler effect is most commonly associated with the changing pitch of a sound as its source moves past you. For example, the pitch of the noise made by a speeding train increases as it approaches you and decreases as it moves away from you. Explain how the Doppler effect is relevant to the study of the universe.

The spectra of light from stars contain dark lines caused by the absorption of certain colours by substances in the star.

As the stars move away our solar system at high speed, the Doppler effect causes the lines to move towards the red end of the spectrum.

As the stars move towards our solar system at high speed, the Doppler effect causes the lines to move towards the blue end of the spectrum.

14. What is cosmic microwave background radiation and why does it exist?

Cosmic microwave background radiation is radiation that was released 380 000 years after the Big Bang.

Human Biology

1. (a) What are the male and female gametes in animals called?

sperm and egg (ovum)

(b) What are the **genotypes** for a male and a female?

male XY female XX

2. DNA is thought to be a twisted ladder structure called a double helix. There are a series of bases located between the two strands of the helix.

(a) Name the four bases.

guanine (G) thymine (T) adenine (A) guanine (G)

(b) Which bases are paired together?

A - T
G - C

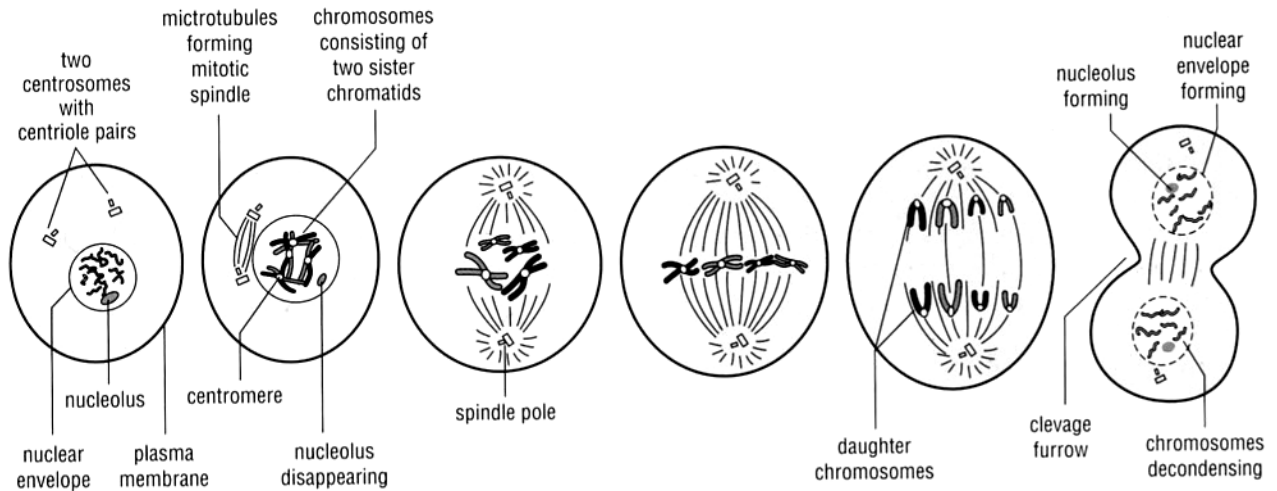
(c) What are the bases made of?

nitrogen - based

3. Define the following terms.

allele	Alternative forms of a gene. e.g. T, t
pure-bred	2 alleles for a gene are the same. e.g. TT, tt
hybrid	2 different alleles. e.g. Tt
homozygous	Pure-bred - alleles are the same. e.g. TT, tt
heterozygous	Hybrid - alleles are different. e.g. Tt
genotype	The genes for a characteristic. e.g. TT, Tt, tt
phenotype	How the organism looks. e.g. Tall, dwarf
sex-linked	Characteristic carried on the X or Y chromosome.
dominant	A gene that "masks" another. e.g. T dominates t
recessive	A gene that doesn't show if the dominant gene is present.
co-dominant	Both characteristics show. e.g. Red + white hair in cattle.
incomplete dominance	New characteristic shows - a "blend". e.g. Pink colour when red + white flowers are crossed.
haploid	Half the normal number of chromosomes.
gene	Information for a specific characteristic.

4. Below is a diagram showing a cell division process.



(a) Is this meiosis or mitosis?

mitosis

(b) Describe what is happening at each stage to the chromosomes.

1. Chromosomes become visible.
2. Chromosomes thicken and duplicate.
3. Chromosomes move towards the equator and spindle fibres form.
4. Chromosomes line up across the equator.
5. Spindle fibres pull the chromosomes apart.
6. Cell starts to "pinch off" to form two daughter cells.

(c) If possible, name each stage.

1 - 3: prophase

4: metaphase

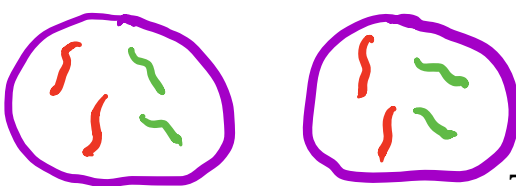
5: anaphase

6: telophase

(d) In which cells does this process occur?

Normal cells - not reproductive cells.

(e) Draw a diagram to show the stages in the process **not** shown above.



Two daughter cells form.
Chromosomes disappear and
interphase occurs.

5. (a) Why do cells undergo mitosis?

Need to replace damaged cells.

General growth - baby into toddler into teenager into adult.

- (b) Why do cells undergo meiosis?

To halve the number of chromosomes in gametes ready for sexual reproduction.

- (c) In what way do the cells produced by these two processes differ?

Mitosis - produces 2 daughter cells with diploid (normal) number of chromosomes.

Meiosis - produces 4 daughter cells with haploid (half) number of chromosomes.

6. Polled cattle do not have horns. This is a dominant phenotype arising from the action of an autosomal gene **P**. Horned cattle are therefore **pp** homozygotes.

- (a) If a polled bull and a polled cow have a horned calf, what are the genotypes of the parents?

♂ Pp × ♀ Pp

Must have "p" present in each parent.

- (b) If the same bull was bred with many horned cows, what would the proportions of polled and horned offspring overall be? **Show full working.**

♂ Pp × ♀ PP

♀ \ ♂	P	p
P	PP	Pp
p	Pp	pp

GENOTYPE

Pp Pp

PP PP

PHENOTYPE

50% poll

50% horned

7. In Andalusian chickens, the black characteristic is **incompletely dominant** to the white-splashed characteristic. The heterozygous (hybrid) condition produces blue chickens.

(a) How is incomplete dominance different to **co-dominance**?

Incomplete - new or blended characteristic occurs. e.g. blue

Co-dominance - both colours show. e.g. black and white would show together.

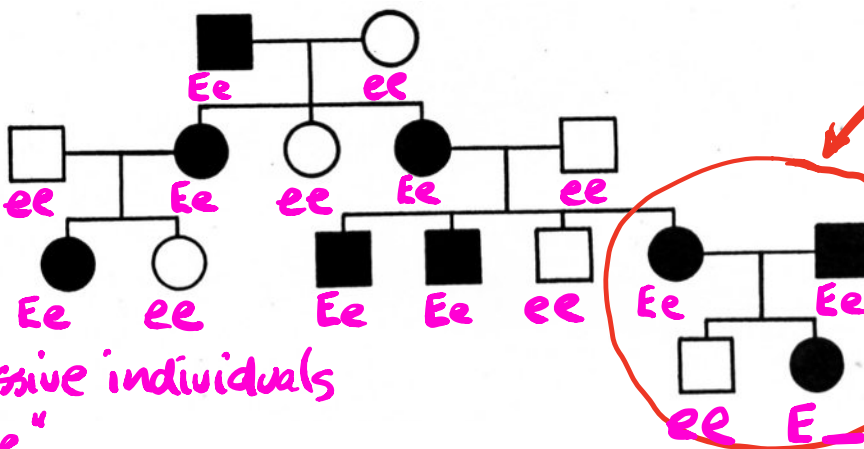
(b) Determine the genotype and phenotype of the F1 generation from a pure-bred black Andalusian crossed with a blue Andalusian.

♂ BB × ♀ BW

♀ \ ♂	B	B
B	BB	BB
W	BW	BW

GENOTYPE BB BB BW BW
 PHENOTYPE 50% black 50% blue

8. The following pedigree shows the occurrence of no ear lobes in a family.



All recessive individuals are "ee".

Parents have the condition but at least one offspring doesn't.

⇒ DOMINANT

Second allele is unknown until children are produced.

(a) Is the characteristic dominant or recessive (according to the pedigree)?

Dominant

(b) Explain why you made that determination.

See note on diagram.

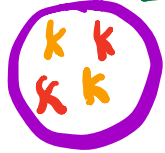
(c) Determine the genotype of every individual in the pedigree.

(d) How many females have no ear lobes?

5 (black circles)

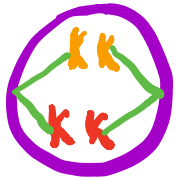
9. Describe, with a simple diagram, the first stages of **meiosis**. Label what is happening at each stage. In your diagrams, use 4 chromosomes as the normal number.

STAGE I



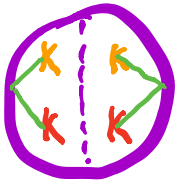
Prophase I

Chromosomes become visible and duplicate.



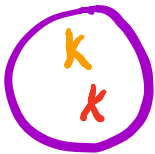
Metaphase I

Line up in pairs across the equator. Spindle fibres form.



Anaphase I

Pairs are separated by the spindle fibres. Cell starts to pinch in (cleave).



Telophase I

2 daughter cells form with a haploid (half) number of chromosomes.

10. (a) How many chromosomes are found in a human skin cell?

46

- (b) How many chromosomes are found in the reproductive cells?

23

- (c) Name the process used to make reproductive cells?

meiosis

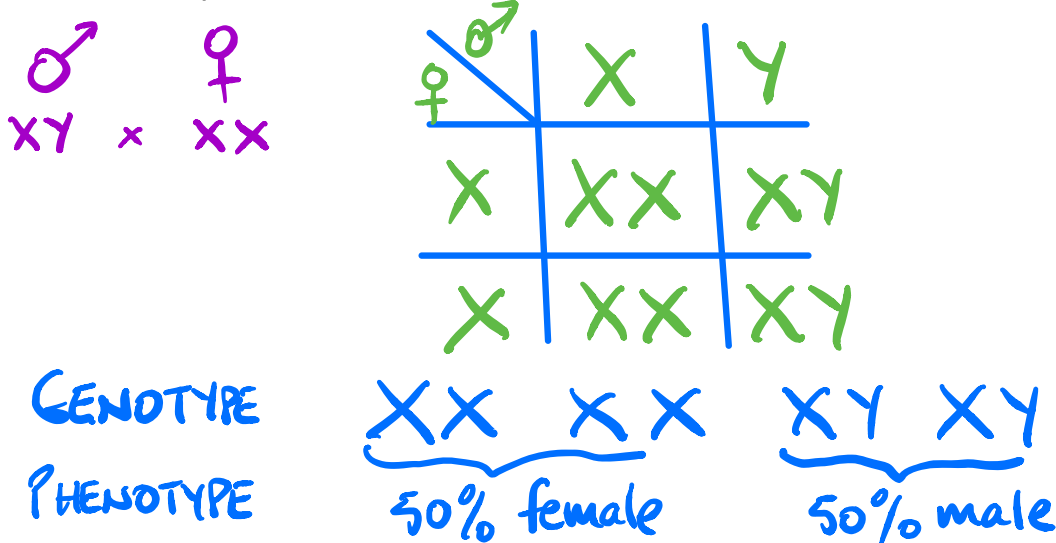
11. (a) What is the difference between the male and female sex chromosome pair?

XY (Y chromosome is very small) vs XX

- (b) Who determines the sex of a child - the mother or father? Explain.

Father - he has the Y chromosome that determines maleness.

- (c) Show, by punnet square, why half of any particular population would be expected to be female.



- (d) A family has had five boys born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

50 : 50 Half of the sperm in the father are Y chromosome.

12. (a) Define the term 'adaptation'.

A change in an organism or species to allow it to better suited to its environment.

A special characteristic that helps an organism survive and reproduce in its environment

- (b) Explain how each of the following examples are adaptations for survival.

- (i) Desert marsupials excrete small quantities of very concentrated urine.

Minimises the amount of moisture lost from the animal.

Expelling only small amounts of concentrated urine helps to retain water in the body and avoid dehydration in the harsh desert climate.

- (ii) Polar bears have a thick fur coat.

Protects the animal from the extreme cold.

The thick fur helps to retain heat and thus avoid hypothermia in the icy environment.

13. The photo below shows the 150 million year old fossilised remains of an *Archaeopteryx*.



The fossilised skeleton of *Archaeopteryx*

- (a) Identify the features that survived the fossilisation process.

Bones and feathers.

- (b) Explain why these features survived whereas others did not.

They did not break down as the organism decayed.

They are hard and not readily broken down and are least subject to decay.

- (c) Explain the scientific importance of this fossil.

Is the link between reptiles (dinosaurs) and birds. Shows how the reptiles evolved feathers.

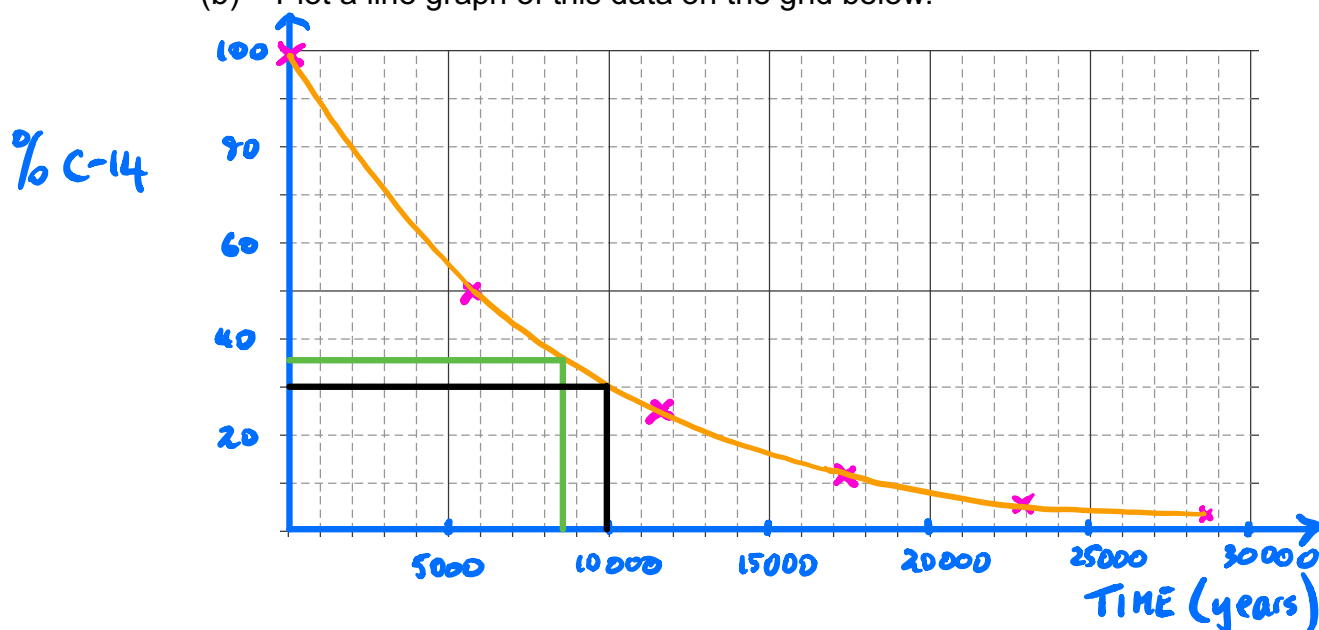
The fossil is an intermediate form between true birds and dinosaurs. It thus provides evidence of evolution.

14. The absolute ages of recent carbonaceous fossils can be determined using radiocarbon dating. The half-life of carbon-14 (a radioactive isotope) is 5730 years. The following table shows how the percentage of C-14 present in a fossil decreases over time.

Time (years)	0	5730	11 460	17 190	22 920	28 650
% C-14 remaining	100	50		12.5	6.25	3.125

(a) Complete the table.

(b) Plot a line graph of this data on the grid below.



(c) The C-14 content of a fossil was found to have decreased by 36% since the organism died. Determine the approximate age of the fossil.

8,500 years

(d) A fossil is approximately 10 000 years old. What percentage of C-14 still remains?

30 %

(e) Explain why it is difficult to date fossils that are more than 60 000 years old using radiocarbon dating.

The amount of C-14 is so low that the count rate for it is lost in the background radiation caused by other naturally-occurring radioactive substances.